

Sharp Plücker Geometry for Three-Copy Werner Distillation

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Abstract

Whether negative-partial-transpose entanglement can remain undistillable is a longstanding problem in quantum information theory. We analyze the first unresolved three-copy Werner endpoint using a sharp dimension-free inequality for complementary partial traces and an optimal exterior-square inequality for orthonormal tripartite vectors. The latter identifies local SWAP-parity statistics with metric data of a decomposable Plücker bivector. Together these inequalities prove endpoint nonnegativity for every positive semidefinite rank-two coefficient operator and for the complete normal rank-two sector in arbitrary finite local dimensions. For genuinely nonnormal operators, an exact crossed-Gram criterion proves nonnegativity when one local output–input support overlap is at most two, including every system with a qubit-sized factor, and when either support plane contains a product ray. Explicit anti-state and rank-boundary families establish optimality of the constants and the rank restriction.

1 Introduction and main results

1.1 Physical setting: Werner states and distillation

Entanglement distillation takes multiple imperfect bipartite states as input and, using only LOCC, produces fewer output pairs with fidelities approaching that of a maximally entangled state. This conversion is a basic primitive of long-distance quantum communication and gives the operational distinction between free and bound entanglement [1–5]. A bipartite state ρ is said to have positive partial transpose (PPT) if $\rho^\Gamma \succeq 0$, where Γ transposes one party in a fixed product basis; positivity of ρ^Γ is independent of the chosen local basis [6, 7]. Every PPT state is undistillable. Whether every negative-partial-transpose (NPT) state is distillable is not known; an NPT counterexample would constitute NPT bound entanglement [8–14]. Such a counterexample would also have consequences for additivity of distillable entanglement and is closely connected with tensor-stable positive maps [15, 16]. Comparison with the larger class of PPT-preserving operations isolates restrictions intrinsic to LOCC beyond partial-transpose negativity [17]. Both single-photon-pair purification assisted by hyperentanglement and the activation of PPT bound entanglement have been realized experimentally [18, 19]; these experiments establish the operational reality of the resource distinction, although neither addresses the NPT finite-copy conjecture studied here.

Werner states form the canonical symmetric test family [20, 21]

$$\rho_\alpha^{(d)} = \frac{I_{d^2} + \alpha F_d}{d^2 + \alpha d}, \quad -1 \leq \alpha \leq 1, \quad (1)$$

on $\mathbb{C}^d \otimes \mathbb{C}^d$, where $F_d(x \otimes y) = y \otimes x$. These states are invariant under $U \otimes U$ conjugation. They are separable precisely when $\alpha \geq -1/d$ and one-copy distillable precisely when $\alpha < -1/2$ [4, 20, 21]. Thus, for $d \geq 3$, the interval $-1/2 \leq \alpha < -1/d$ consists of NPT states that are not one-copy distillable. The recovery of operational quantum resources from two-qutrit Werner states by local filtering has also been demonstrated experimentally [22].

A state ρ is k -copy distillable if an LOCC protocol can map $\rho^{\otimes k}$, with nonzero probability, to an entangled two-qubit state. The Schmidt rank of a bipartite vector is the number of nonzero terms in its Schmidt decomposition. Equivalently, there is a unit vector ψ of Schmidt rank at most two such that

$$\langle \psi, (\rho^\Gamma)^{\otimes k} \psi \rangle < 0$$

[2–4]. The finite-copy criterion for (1) is most transparent after vectorization. Write

$$|\text{vec } C\rangle = \sum_{a,b} C_{a,b} |a\rangle_A |b\rangle_B.$$

The Schmidt rank of this vector across $A : B$ equals $\text{rank } C$. Because $F_d^\Gamma = |\Omega_d\rangle\langle\Omega_d|$, with $|\Omega_d\rangle = \sum_j |jj\rangle$, direct contraction gives [23–26]

$$\langle \text{vec } C | [(\rho_\alpha^{(d)})^\Gamma]^{\otimes k} | \text{vec } C \rangle = \frac{q_k(\alpha, C)}{(d^2 + \alpha d)^k}, \quad q_k(\alpha, C) = \sum_{J \subseteq [k]} \alpha^{|J|} \|\text{Tr}_J C\|_2^2. \quad (2)$$

Here and below, $[k] = \{1, \dots, k\}$. Consequently, $\rho_\alpha^{(d)}$ is k -copy distillable if and only if $q_k(\alpha, C) < 0$ for some C of rank at most two. This criterion is recalled with its index contraction in Appendix A.

Four recent works have established the exact two-copy threshold [26–29]. The three-copy endpoint $\alpha = -1/2$ is the first unresolved layer beyond that threshold. For an operator C on $\mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \mathcal{H}_3$, it is governed by

$$q_3(C) = \|C\|_2^2 - \frac{1}{2} \sum_{i=1}^3 \|\text{Tr}_i C\|_2^2 + \frac{1}{4} \sum_{1 \leq i < j \leq 3} \|\text{Tr}_{ij} C\|_2^2 - \frac{1}{8} |\text{Tr } C|^2. \quad (3)$$

The tensor factors in $\mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \mathcal{H}_3$ label the three resource copies; the physical bipartition remains $A_1 A_2 A_3 : B_1 B_2 B_3$. Thus “tripartite” below describes the coefficient-space tensor structure, not a third communicating party. The alternating weights in (3) form a signed ledger of how much Hilbert–Schmidt coherence remains after zero, one, two, or all three copy slots are contracted. The two-copy theorem cannot simply be reapplied after tracing out one slot, because a partial trace of a rank-two operator need not remain rank two. At three copies the reductions must instead be controlled simultaneously. Estimating the cuts independently forgets that they all originate from the same two-dimensional global support; the Plücker step restores precisely this common-origin constraint.

We use the following sesquilinear polarization throughout:

$$b_3(X, Y) = \sum_{J \subseteq \{1,2,3\}} \left(-\frac{1}{2}\right)^{|J|} \text{Tr} \left[(\text{Tr}_J X)^\dagger \text{Tr}_J Y \right], \quad q_3(X) = b_3(X, X). \quad (4)$$

Here Tr_J traces out the tensor factors indexed by J , with $\text{Tr}_\emptyset C = C$. The local spaces need not have equal dimensions in any of the results below. The unrestricted endpoint asks whether (3) is nonnegative on every C of rank at most two. Here C is the coefficient matrix of a distillation test vector and need not be a density operator. We first prove nonnegativity on the positive

semidefinite cone. Combining the two spectral signs then closes the complete normal rank-two sector. Polarization beyond normality isolates one exact crossed-Gram determinant. We prove two broad closures of that determinant: one local output–input support overlap is at most two, or either global support plane contains a product ray. The normal result also gives the common initial–final two-plane case directly. Appendix C treats the more specialized local-traceless and crossed product-code sectors.

Throughout, all Hilbert spaces are finite-dimensional and complex. For a Hilbert space $\mathcal{H}$, $\mathcal{L}(\mathcal{H})$ denotes its linear operators and I the identity operator. The symbols † , T , and an overline denote, respectively, the Hilbert-space adjoint, matrix transpose, and entrywise complex conjugation in the fixed product bases; $\operatorname{Re} z$ is the real part of a scalar z . The orders $A \succeq B$ and $A \preceq B$ are the Loewner orders, and $\operatorname{ran} A$ is the range of A . The identity superoperator is denoted by id , and $\operatorname{SR}(\psi)$ denotes the Schmidt rank of a bipartite vector.

The inner product $\langle \cdot, \cdot \rangle$ is conjugate-linear in its first argument. All partial traces are the standard unnormalized partial traces. For $J \subseteq [k]$, $\bar{J} = J^c = [k] \setminus J$. In the tripartite setting, $\bar{i} = \{1, 2, 3\} \setminus \{i\}$, $d_i = \dim \mathcal{H}_i$, and

$$C_i := \operatorname{Tr}_{\bar{i}} C, \quad C_{\bar{i}} := \operatorname{Tr}_i C.$$

Thus the subscript of Tr_J records the factors that are traced out, whereas the subscript of a reduced operator records the factors that remain. Finally, $|X| = (X^\dagger X)^{1/2}$, and

$$\|X\|_p = (\operatorname{Tr} |X|^p)^{1/p} \quad (1 \leq p < \infty)$$

denotes the Schatten norm. In particular, $\|\cdot\|_1$ is the trace or nuclear norm, $\|\cdot\|_2$ is the Hilbert–Schmidt norm, $\|\cdot\|_\infty$ is the operator norm, and $\langle A, B \rangle_{\text{HS}} := \operatorname{Tr}(A^\dagger B)$.

1.2 Three sharp inequalities

We present three theorems as the main results in this paper. The first one compresses each bipartite cut without a dimension or rank penalty. Then the second one restores the common two-plane compatibility across the three cuts, and the third statement converts that geometry into the required distillation inequality. Their composition is the three-copy mechanism.

Partial-trace norm inequalities supply the dimension-free input to our argument. Their development has a substantial history. Audenaert proved a Schatten- q inequality for positive operators, Rastegin developed bounds for broader classes of unitarily invariant norms, and Costa Rico and Wolf recently extended the subadditivity framework to nonnormal matrices [30–32]. Our estimate removes both positivity and rank restrictions by measuring the original operator in nuclear norm, while retaining a dimension-free joint quadratic bound. This rank-unrestricted estimate is unprecedented in the existing partial-trace literature. More precisely, Proposition 11 of [32] gives, for rank-one M ,

$$\|\operatorname{Tr}_{\mathcal{K}} M\|_2^2 + \|\operatorname{Tr}_{\mathcal{L}} M\|_2^2 \leq \|M\|_2^2 + |\operatorname{Tr} M|^2,$$

whereas [29] prove, for rank $M \leq r$,

$$\|\operatorname{Tr}_{\mathcal{K}} M\|_2^2 + \|\operatorname{Tr}_{\mathcal{L}} M\|_2^2 \leq r\|M\|_2^2 + r^{-1}|\operatorname{Tr} M|^2.$$

The following estimate complements these rank-constrained Hilbert–Schmidt bounds by replacing the rank parameter with the nuclear norm.

Theorem 1.1 (Sharp quadratic partial-trace inequality). *For every $X \in \mathcal{L}(\mathcal{K} \otimes \mathcal{L})$,*

$$\|\operatorname{Tr}_{\mathcal{K}} X\|_2^2 + \|\operatorname{Tr}_{\mathcal{L}} X\|_2^2 \leq \|X\|_1^2 + |\operatorname{Tr} X|^2. \quad (5)$$

Neither coefficient on the right can be decreased while the other is held fixed.

Nuclear norms also enter entanglement theory through projective cross-norm and realignment criteria [33, 34]. The common feature of realignment and Theorem 1.1 is that $\|\cdot\|_1$ retains the full singular-value aggregate that a Hilbert–Schmidt estimate may discard.

The geometric consequence acts on an orthonormal pair u, v in $\mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \mathcal{H}_3$. On two copies, put $P_i^\pm = (I \pm F_i)/2$, where F_i swaps the two copies of $\mathcal{H}_i$. These three swaps commute. If $S \subseteq \{1, 2, 3\}$, let

$$\Pi_S = \prod_{i \in S} P_i^- \prod_{i \notin S} P_i^+,$$

and define

$$a(u) = \sum_{|S|=2} \|\Pi_S(u \otimes u)\|^2, \quad (6)$$

$$p_2(u, v) = \sum_{|S|=2} \|\Pi_S(u \otimes v)\|^2, \quad (7)$$

$$p_3(u, v) = \|P_1^- P_2^- P_3^-(u \otimes v)\|^2. \quad (8)$$

The two copies in these definitions are auxiliary replicas of the coefficient-space rays. Thus $\|\Pi_S(u \otimes v)\|^2$ is the joint probability of obtaining antisymmetric local SWAP parity precisely at the sites in S . Accordingly, $a(u)$ is the total two-minus probability for two copies of u , while $p_2(u, v)$ and $p_3(u, v)$ are the two-minus and three-minus probabilities for the ordered pair $u \otimes v$. Symmetric and antisymmetric two-copy projectors also underlie multipartite-concurrence constructions and scalable parity-based entanglement estimation [35, 36]. Here their joint three-site distribution is constrained by one decomposable Plücker bivector. Such parity probabilities, and the subsystem purities to which they reduce, can be measured by two-copy interference [37–42]. For qubits, P_i^- and P_i^+ are respectively the singlet and triplet projectors. Recent trapped-ion experiments identify the corresponding Bell-sampling triplet statistics with quantum shadow enumerators [42].

Theorem 1.2 (Sharp exterior-square (Plücker-coordinate) inequality). *Every orthonormal tripartite pair satisfies*

$$3p_3(u, v) - p_2(u, v) \leq \sqrt{a(u)a(v)}. \quad (9)$$

The constant is optimal in every set of local dimensions containing a $2 \times 2 \times 2$ subspace.

The term “Plücker” records that $u \wedge v$ is a decomposable bivector, hence a point of the Grassmannian in its Plücker embedding [43–45]. Its fully locally antisymmetric component controls p_3 , while $a(u), a(v)$ are sums of squared second-compound norms. Theorem 1.2 retains the decomposability constraint that is lost if the local swap-sector weights are treated as arbitrary probabilities. The theorem uses the metric geometry of the Plücker embedding. Each cut sees a different projection of the same oriented two-area $u \wedge v$.

Theorem 1.3 (Positive rank-two consequence). *If $C \succeq 0$ and $\text{rank } C \leq 2$, then*

$$\mathcal{S}(C) \geq 0, \quad (10)$$

$$q_3(C) \geq \frac{2 \text{Tr } C^2 - (\text{Tr } C)^2}{8} \geq 0, \quad (11)$$

where

$$\mathcal{S}(C) = 3\|C\|_2^2 + \sum_i \|\text{Tr}_{\bar{i}} C\|_2^2 - 2 \sum_i \|\text{Tr}_i C\|_2^2. \quad (12)$$

Here $\text{Tr}_{\bar{i}}$ leaves subsystem i , whereas Tr_i leaves the complementary pair.

For a normalized state, $\mathcal{S}(C) \geq 0$ is equivalently

$$3S_L(C) + \sum_i S_L(C_i) \leq 2 \sum_i S_L(C_{\bar{i}}), \quad S_L(\rho) = 1 - \text{Tr } \rho^2. \quad (13)$$

The linear entropy is a directly measurable second-order impurity, distinct from the von Neumann entropy. Equation (13) is therefore a purity constraint, separate from the usual strong subadditivity theorem. It belongs to the family of quantum shadow and sector-length inequalities [46–50]. Wyderka and Gühne proved (13) for all three-qubit states [50]; Theorem 1.3 allows arbitrary local dimensions under the rank-two hypothesis.

The proof begins with the exact two-vector reduction in Section 2 and the sharp projection lemma in Section 3. Sections 4 and 5 establish the two main inequalities. Section 6 derives their positive, normal, and common-plane consequences. Section 7 identifies the remaining nonnormal determinant and proves two broad closures. Section 8 constructs equality families and rank boundaries; Section 9 gives the operational interpretation. The appendices contain the contraction identities and two additional closed sectors.

2 Exact two-vector reduction

We first separate the two spectral rays and express every mixed term in local SWAP-parity variables. Let

$$C = \lambda P_u + \mu P_v, \quad \lambda, \mu \geq 0, \quad \langle u, v \rangle = 0, \quad (14)$$

where $P_w = |w\rangle\langle w|$. For a unit tripartite vector w , write $\rho_S^w = \text{Tr}_{\bar{S}} P_w$, and set

$$s_w = \sum_{i=1}^3 \text{Tr}[(\rho_i^w)^2], \quad e_i(w) = 1 - \text{Tr}[(\rho_i^w)^2]. \quad (15)$$

Thus $e_i(w)$ is the linear entropy of entanglement across the pure bipartition $i : \bar{i}$. Expanding the local swap projectors gives

$$a(w) = \frac{1}{4}(3 - s_w) = \frac{1}{4} \sum_i e_i(w). \quad (16)$$

For an orthonormal pair u, v , define the marginal overlaps

$$x_i = \text{Tr}(\rho_i^u \rho_i^v), \quad y_{ij} = \text{Tr}(\rho_{ij}^u \rho_{ij}^v).$$

The Walsh expansion is the discrete Fourier transform on $\mathbb{Z}_2^3$. Expanding the three commuting parity projectors and using the SWAP identity gives, for every $S \subseteq \{1, 2, 3\}$,

$$\|\Pi_S(u \otimes v)\|^2 = \frac{1}{8} \sum_{T \subseteq \{1, 2, 3\}} (-1)^{|S \cap T|} \text{Tr}(\rho_T^u \rho_T^v). \quad (17)$$

The empty-set character is one, while the full-set character is $\text{Tr}(P_u P_v) = |\langle u, v \rangle|^2 = 0$. Summing the characters for the all-minus outcome and for the three outcomes with exactly two minus signs therefore gives

$$p_3 = \frac{1}{8} \left(1 - \sum_i x_i + \sum_{i < j} y_{ij} \right), \quad (18)$$

$$p_2 = \frac{1}{8} \left(3 - \sum_i x_i - \sum_{i < j} y_{ij} \right). \quad (19)$$

In particular,

$$4(p_2 - 3p_3) = \sum_i x_i - 2 \sum_{i < j} y_{ij}. \quad (20)$$

The following algebraic identity does not require positivity, normality, or a rank assumption. For every tripartite operator C , direct coefficient comparison between (3) and (12) gives

$$q_3(C) = \frac{1}{4}\mathcal{S}(C) + \frac{1}{8}(2\|C\|_2^2 - |\operatorname{Tr} C|^2). \quad (21)$$

Proposition 2.1 (Exact positive spectral bridge). *For $C = \lambda P_u + \mu P_v$ as above,*

$$\mathcal{S}(C) = 4 \left[\lambda^2 a(u) + \mu^2 a(v) + 2\lambda\mu(p_2(u, v) - 3p_3(u, v)) \right]. \quad (22)$$

Proof. For a pure state, complementary purities agree. Hence

$$\mathcal{S}(P_w) = 3 - \sum_i \operatorname{Tr}[(\rho_i^w)^2] = 4a(w).$$

The cross coefficient in $\mathcal{S}(\lambda P_u + \mu P_v)$ is

$$2\lambda\mu \left(\sum_i x_i - 2 \sum_{i < j} y_{ij} \right),$$

because $\operatorname{Tr}(P_u P_v) = 0$. Equation (20) proves (22). $\square$

Remark 2.2. Theorem 1.2 immediately turns (22) into a perfect square:

$$\frac{1}{4}\mathcal{S}(C) \geq (\lambda\sqrt{a(u)} - \mu\sqrt{a(v)})^2.$$

The remaining sections establish the Plücker estimate without assuming local dimensions.

3 The double-antisymmetric projection

The sharp partial-trace constant comes from one projection problem. Let $\mathcal{K}, \mathcal{L}$ be finite-dimensional Hilbert spaces. On $(\mathcal{K} \otimes \mathcal{L})^{\otimes 2}$, let $F_{\mathcal{K}}$ and $F_{\mathcal{L}}$ be the local copy swaps and put

$$Q = P_{\mathcal{K}}^- P_{\mathcal{L}}^-, \quad P_{\mathcal{K}}^- = \frac{I - F_{\mathcal{K}}}{2}, \quad P_{\mathcal{L}}^- = \frac{I - F_{\mathcal{L}}}{2}.$$

For an orthogonal projection P , the variational quantity

$$\sup_{\substack{\|\psi\|=1 \\ \operatorname{SR}(\psi) \leq k}} \langle \psi, P\psi \rangle$$

is its $S(k)$ -operator norm [51, 52]. The following lemma determines the $S(2)$ -norm of Q . It is the sharp form of the half-property studied by Pankowski *et al.* and is the geometric input of the recent two-copy solution [12, 27]. Fu, Gao, and Park state their Theorem 2.4 for equal local dimensions; the proof below gives the rectangular $\mathcal{K} \otimes \mathcal{L}$ form needed here.

Lemma 3.1 (Sharp double-antisymmetric projection). *If ψ is a unit vector of Schmidt rank at most two across the copy cut*

$$(\mathcal{K} \otimes \mathcal{L})_{\text{copy } 1} : (\mathcal{K} \otimes \mathcal{L})_{\text{copy } 2},$$

then

$$\langle \psi, Q\psi \rangle \leq \frac{1}{2}. \quad (23)$$

If $\dim \mathcal{K}, \dim \mathcal{L} \geq 2$, the constant $1/2$ is optimal.

Proof. This is the sharp projection statement used by Fu, Gao, and Park [27, Theorem 2.4]; we include a self-contained proof. If $\dim \mathcal{K} < 2$ or $\dim \mathcal{L} < 2$, then one local antisymmetric projector vanishes, so $Q = 0$ and the assertion is immediate. Hence assume $\dim \mathcal{K}, \dim \mathcal{L} \geq 2$. For a unit $\phi \in \text{ran } Q$, the full copy swap $F_{\mathcal{K}}F_{\mathcal{L}}$ fixes ϕ . Takagi factorization therefore gives

$$\phi = \sum_j s_j e_j \otimes e_j, \quad s_1 \geq s_2 \geq \dots \geq 0, \quad \sum_j s_j^2 = 1.$$

The largest squared overlap of ϕ with a unit vector of Schmidt rank at most two is $s_1^2 + s_2^2$. Because Q is an orthogonal projection, taking the two suprema in either order gives

$$\begin{aligned} \sup_{\substack{\|\psi\|=1 \\ \text{SR}(\psi) \leq 2}} \langle \psi, Q\psi \rangle &= \sup_{\substack{\|\psi\|=1 \\ \text{SR}(\psi) \leq 2}} \sup_{\substack{\phi \in \text{ran } Q \\ \|\phi\|=1}} |\langle \phi, \psi \rangle|^2 \\ &= \sup_{\substack{\phi \in \text{ran } Q \\ \|\phi\|=1}} (s_1(\phi)^2 + s_2(\phi)^2). \end{aligned} \quad (24)$$

Fix ϕ , put $t = s_1^2 + s_2^2$, and define

$$\eta = t^{-1/2}(s_1 e_1 \otimes e_1 + s_2 e_2 \otimes e_2) = c_1 e_1 \otimes e_1 + c_2 e_2 \otimes e_2.$$

Identify $e_j \in \mathcal{K} \otimes \mathcal{L}$ with a rectangular matrix E_j . Since η is symmetric under the full copy swap,

$$\langle \eta, Q\eta \rangle = \frac{1}{2}(1 - \langle \eta, F_{\mathcal{L}}\eta \rangle).$$

The remaining expectation is

$$\begin{aligned} \langle \eta, F_{\mathcal{L}}\eta \rangle &= c_1^2 \text{Tr}[(E_1^\dagger E_1)^2] + c_2^2 \text{Tr}[(E_2^\dagger E_2)^2] + 2c_1 c_2 \text{Re} \text{Tr}(E_1^\dagger E_2 E_1^\dagger E_2) \\ &\geq (c_1 \|E_1\|_4^2 - c_2 \|E_2\|_4^2)^2 \geq 0. \end{aligned}$$

Indeed,

$$|\text{Tr}(E_1^\dagger E_2 E_1^\dagger E_2)| \leq \|E_1^\dagger E_2\|_2^2 \leq \|E_1\|_4^2 \|E_2\|_4^2$$

by Schatten Hölder. Thus $\|Q\eta\|^2 = \langle \eta, Q\eta \rangle \leq 1/2$. Since $Q\phi = \phi$,

$$t = |\langle \eta, \phi \rangle|^2 = |\langle Q\eta, \phi \rangle|^2 \leq \|Q\eta\|^2 \leq \frac{1}{2}.$$

Taking the supremum in (24) proves the bound. For optimality, choose orthonormal local vectors $|1\rangle, |2\rangle$ and let

$$\psi = \frac{|11\rangle_{\text{copy } 1} |22\rangle_{\text{copy } 2} + |22\rangle_{\text{copy } 1} |11\rangle_{\text{copy } 2}}{\sqrt{2}}.$$

It has Schmidt rank two across the copy cut and $\langle \psi, Q\psi \rangle = 1/2$. $\square$

4 Proof of the quadratic partial-trace inequality

The SVD transfers the projection bound from two-copy vectors to an arbitrary bipartite operator.

Proof of Theorem 1.1. Take a singular-value decomposition

$$X = \sum_{\alpha=1}^r s_{\alpha} |u_{\alpha}\rangle\langle v_{\alpha}|, \quad s_{\alpha} > 0,$$

where each of the two vector families is orthonormal. Put

$$X_{\alpha} = |u_{\alpha}\rangle\langle v_{\alpha}|, \quad \Phi(Z) = (\text{Tr}_{\mathcal{K}} Z, \text{Tr}_{\mathcal{L}} Z).$$

The two-component target of Φ carries the direct-sum Hilbert–Schmidt inner product. With $F = F_{\mathcal{K}} F_{\mathcal{L}}$, the operator

$$K = F_{\mathcal{K}} + F_{\mathcal{L}} - I = F - 4Q$$

satisfies $QF = FQ = Q$. The swap trick gives

$$\begin{aligned} \langle \Phi(|u\rangle\langle v|), \Phi(|s\rangle\langle t|) \rangle_{\text{HS}} - \overline{\text{Tr}(|u\rangle\langle v|)} \text{Tr}(|s\rangle\langle t|) \\ = \langle u \otimes t, K(v \otimes s) \rangle. \end{aligned} \quad (25)$$

For example, the $\text{Tr}_{\mathcal{K}}$ term on the left equals $\langle u \otimes t, F_{\mathcal{L}}(v \otimes s) \rangle$.

Define

$$\gamma_{\alpha\beta} = \langle \Phi(X_{\alpha}), \Phi(X_{\beta}) \rangle_{\text{HS}} - \overline{\text{Tr } X_{\alpha}} \text{Tr } X_{\beta}.$$

On the diagonal, (25) gives

$$\gamma_{\alpha\alpha} = 1 - 4\|Q(u_{\alpha} \otimes v_{\alpha})\|^2 \leq 1.$$

For $\alpha \neq \beta$, the full-swap contribution vanishes by orthogonality. Set

$$a = u_{\alpha} \otimes v_{\beta}, \quad b = u_{\beta} \otimes v_{\alpha}, \quad z = \langle a, Qb \rangle.$$

Equivalently,

$$\gamma_{\alpha\beta} = \langle a, (I - 4Q)b \rangle = -4\langle a, Qb \rangle = -4z,$$

because $\langle a, b \rangle = 0$. The vectors a, b are orthonormal, and $\eta_{\theta} = (a + e^{i\theta}b)/\sqrt{2}$ has Schmidt rank at most two across the copy cut. Choose θ so that the cross term is $|z|$, and set

$$p = \langle a, Qa \rangle, \quad q = \langle b, Qb \rangle.$$

Lemma 3.1 and Cauchy–Schwarz in the semidefinite inner product induced by Q give, respectively,

$$p + q + 2|z| \leq 1, \quad 2|z| \leq 2\sqrt{pq} \leq p + q.$$

Consequently $|z| \leq 1/4$, and hence $\text{Re } \gamma_{\alpha\beta} \leq 1$ also off the diagonal. It follows that

$$\begin{aligned} & \| \text{Tr}_{\mathcal{K}} X \|_2^2 + \| \text{Tr}_{\mathcal{L}} X \|_2^2 - | \text{Tr } X |^2 \\ &= \sum_{\alpha, \beta} s_{\alpha} s_{\beta} \text{Re } \gamma_{\alpha\beta} \leq \left(\sum_{\alpha} s_{\alpha} \right)^2 = \|X\|_1^2. \end{aligned}$$

The double sum is real because $\gamma_{\beta\alpha} = \overline{\gamma_{\alpha\beta}}$.

For a rank-one projector onto a product of unit vectors, the two sides of (5) equal two; with the coefficient of $\|X\|_1^2$ fixed, this forces the coefficient of $| \text{Tr } X |^2$ to be at least one. Conversely, take $X = |a\rangle\langle a| \otimes |b\rangle\langle b'|$ with a, b, b' unit and $\langle b, b' \rangle = 0$. Its left side and $\|X\|_1^2$ both equal one, while $\text{Tr } X = 0$, forcing the coefficient of $\|X\|_1^2$ to be at least one. $\square$

5 Exterior-square control and the sharp Plücker theorem

We now retain the second compound of each reduced transition operator. Use the standard Hilbert exterior-power normalization

$$e_i \wedge e_j \mapsto \frac{e_i \otimes e_j - e_j \otimes e_i}{\sqrt{2}} \quad (i < j).$$

Thus the wedges of an orthonormal basis are orthonormal. The symbol $\wedge^2 X$ denotes the induced map on the second exterior power; if $\sigma_1(X) \geq \sigma_2(X) \geq \dots$ are the singular values of X , then the singular values of $\wedge^2 X$ are $\sigma_i(X)\sigma_j(X)$, $i < j$.

Lemma 5.1 (Transition exterior-square bound). *Let u, v be normalized vectors in $\mathcal{H}_{\bar{k}} \otimes \mathcal{H}_k$, and put*

$$X_k = \text{Tr}_k |u\rangle\langle v|.$$

Then

$$\|X_k\|_1^2 - \|X_k\|_2^2 \leq \sqrt{e_k(u)e_k(v)}. \quad (26)$$

Proof. Flatten u, v across $\bar{k} : k$ into rectangular matrices $U, V : \mathcal{H}_k \rightarrow \mathcal{H}_{\bar{k}}$. Then $X_k = UV^\dagger$ and $\|U\|_2 = \|V\|_2 = 1$. With the stated vectorization convention,

$$U^\dagger U = (\rho_k^u)^T, \quad V^\dagger V = (\rho_k^v)^T;$$

the transpose does not change either spectrum or purity. The singular values of $\wedge^2 X_k$ are the pairwise products of those of X_k , so

$$\|X_k\|_1^2 - \|X_k\|_2^2 = 2\|\wedge^2 X_k\|_1.$$

These standard compound-matrix identities hold for rectangular maps as well [53]. Functoriality of exterior powers and Schatten Hölder imply

$$\|\wedge^2 X_k\|_1 \leq \|\wedge^2 U\|_2 \|\wedge^2 V\|_2.$$

Finally,

$$\|\wedge^2 U\|_2^2 = \frac{1}{2} \left(\|U\|_2^4 - \|U^\dagger U\|_2^2 \right) = \frac{1}{2} \left(1 - \text{Tr}[(\rho_k^u)^2] \right) = \frac{1}{2} e_k(u),$$

and similarly for V . Combining these identities proves (26). $\square$

Proof of Theorem 1.2. For each $k \in \{1, 2, 3\}$, let $\{i, j, k\} = \{1, 2, 3\}$, and consider

$$X_k = \text{Tr}_k |u\rangle\langle v| \in \mathcal{L}(\mathcal{H}_i \otimes \mathcal{H}_j).$$

The transition-operator identity

$$\|\text{Tr}_S |u\rangle\langle v|\|_2^2 = \text{Tr}(\rho_S^u \rho_S^v) \quad (27)$$

and $u \perp v$ give

$$\|X_k\|_2^2 = x_k, \quad \|\text{Tr}_i X_k\|_2^2 = y_{ik}, \quad \|\text{Tr}_j X_k\|_2^2 = y_{jk}, \quad \text{Tr } X_k = 0.$$

Theorem 1.1 and Lemma 5.1 therefore yield the cutwise estimate

$$y_{ik} + y_{jk} - x_k \leq \sqrt{e_k(u)e_k(v)}. \quad (28)$$

Summing over k counts every y_{ij} twice. Cauchy–Schwarz then gives

$$2 \sum_{i < j} y_{ij} - \sum_k x_k \leq \sum_k \sqrt{e_k(u) e_k(v)} \leq \sqrt{\sum_k e_k(u) \sum_k e_k(v)}.$$

Using (19) and (16),

$$3p_3 - p_2 = \frac{1}{4} \left(2 \sum_{i < j} y_{ij} - \sum_k x_k \right) \leq \sqrt{a(u) a(v)}.$$

Proposition 8.2 below supplies equality examples. $\square$

Remark 5.2. The nuclear norm in Theorem 1.1 is not replaced by a rank bound. Instead, Lemma 5.1 keeps the second compound matrix of $X_k = U_k V_k^\dagger$. Only after obtaining the three cutwise geometric means in (28) is the ordinary Cauchy–Schwarz inequality applied.

6 Rank-two consequences

The Plücker estimate closes the positive spectral sector through the exact bridge (22).

Proof of Theorem 1.3. Equations (22) and (9) give the stronger square

$$\frac{1}{4} \mathcal{S}(C) \geq (\lambda \sqrt{a(u)} - \mu \sqrt{a(v)})^2 \geq 0. \quad (29)$$

Moreover,

$$\|C\|_2^2 = \lambda^2 + \mu^2, \quad \text{Tr } C = \lambda + \mu,$$

so

$$2\|C\|_2^2 - |\text{Tr } C|^2 = (\lambda - \mu)^2 \geq 0.$$

Substitution into (21) proves (11). $\square$

Recall that C is normal when $C^\dagger C = C C^\dagger$, equivalently when it admits an orthonormal spectral basis. Theorem 1.3 controls the same-phase positive spectral combination. The opposite-sign theorem in [24, Proposition 3] controls the other real extreme of the polarized cross term; together, the two bounds cover arbitrary relative phases. In the present normalization, the opposite-sign result states

$$q_3(\alpha P_u - \beta P_v) \geq 0 \quad (u \perp v, \alpha, \beta > 0).$$

Corollary 6.1 (Normal rank-two endpoint). *Every normal tripartite operator C of rank at most two obeys $q_3(C) \geq 0$.*

Proof. Write

$$C = z_1 P_u + z_2 P_v, \quad u \perp v,$$

and use the sesquilinear polarization b_3 of q_3 . Put

$$A = q_3(P_u), \quad B = q_3(P_v), \quad Z = b_3(P_u, P_v) \in \mathbb{R}.$$

Theorem 1.3, applied to every positive spectral pair, gives

$$\alpha^2 A + \beta^2 B + 2\alpha\beta Z \geq 0 \quad (\alpha, \beta \geq 0),$$

and hence $Z \geq -\sqrt{AB}$. Costa Rico's opposite-sign theorem gives

$$\alpha^2 A + \beta^2 B - 2\alpha\beta Z \geq 0 \quad (\alpha, \beta > 0),$$

and hence $Z \leq \sqrt{AB}$. Thus $|Z| \leq \sqrt{AB}$, and

$$q_3(C) = |z_1|^2 A + |z_2|^2 B + 2 \operatorname{Re}(\overline{z_1} z_2) Z \geq (|z_1| \sqrt{A} - |z_2| \sqrt{B})^2 \geq 0.$$

□

Corollary 6.2 (Common initial-final two-plane). *If*

$$\dim(\operatorname{ran} C + \operatorname{ran} C^\dagger) \leq 2, \tag{30}$$

then $q_3(C) \geq 0$.

Proof. Let E be a subspace of dimension at most two containing both ranges, and let P_E be its orthogonal projection. Then

$$C = P_E C P_E.$$

Writing

$$H = \frac{C + C^\dagger}{2}, \quad K = \frac{C - C^\dagger}{2i},$$

we see that the Hermitian operators H, K are supported on E and therefore have rank at most two. Every partial trace preserves adjoints, and for Hermitian X, Y ,

$$\|X + iY\|_2^2 = \|X\|_2^2 + \|Y\|_2^2$$

because $\operatorname{Tr}(XY - YX) = 0$. Applying this identity to every term of (3) gives $q_3(C) = q_3(H) + q_3(K)$. Both summands are nonnegative by Corollary 6.1. □

7 The residual nonnormal obstruction

7.1 The crossed-Gram criterion

Normality aligns the input and output directions within one spectral two-plane. Once this alignment is lost, the new obstruction is the crossed pairing of two output rays with two input rays. The singular-value decomposition isolates all of this genuinely nonnormal interference in one 2×2 determinant.

Proposition 7.1 (Exact crossed-Gram reduction). *Let u_1, u_2 and v_1, v_2 be orthonormal pairs and put $A_r = |u_r\rangle\langle v_r|$. Then*

$$q_3(z_1 A_1 + z_2 A_2) \geq 0 \quad \text{for all } z_1, z_2 \in \mathbb{C}$$

if and only if

$$G(u_1, u_2; v_1, v_2) = \begin{pmatrix} \frac{q_3(A_1)}{b_3(A_1, A_2)} & b_3(A_1, A_2) \\ b_3(A_1, A_2) & q_3(A_2) \end{pmatrix} \succeq 0. \tag{31}$$

Equivalently, the remaining scalar inequality is

$$|b_3(A_1, A_2)|^2 \leq q_3(A_1) q_3(A_2). \tag{32}$$

Thus $q_3(C) \geq 0$ for every $\operatorname{rank} C \leq 2$ if and only if (32) holds for every two orthonormal pairs.

Proof. Polarization gives

$$q_3(z_1 A_1 + z_2 A_2) = \begin{pmatrix} \bar{z}_1 & \bar{z}_2 \end{pmatrix} G(u_1, u_2; v_1, v_2) \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}.$$

This quadratic form is nonnegative for all coefficients exactly when $G \succeq 0$, which is equivalent to (32) because its diagonal entries are positive. Indeed, on two copies of the tripartite space define

$$R = \bigotimes_{i=1}^3 \left(I - \frac{1}{2} F_i \right). \quad (33)$$

Every factor in (33) has eigenvalues $1/2$ and $3/2$, so $R \succeq I/8$. The swap trick yields

$$q_3(A_r) = \langle u_r \otimes v_r, R(u_r \otimes v_r) \rangle, \quad (34)$$

$$b_3(A_1, A_2) = \langle u_1 \otimes v_2, R(u_2 \otimes v_1) \rangle. \quad (35)$$

Hence $q_3(A_r) \geq 1/8$. Finally, every rank-two operator has a singular-value decomposition with precisely the two orthonormal families appearing above, and arbitrary phases can be absorbed into one of those families. $\square$

For a fixed coefficient vector (z_1, z_2) , negativity of $q_3(z_1 A_1 + z_2 A_2)$ implies $G \not\succeq 0$, but $G \not\succeq 0$ need not imply negativity for that prescribed vector. It does, however, guarantee negativity for some coefficient vector within the same four rays; equivalently, this occurs when (32) fails.

The crossed ordering carries the entire obstruction. Let $U, V : \mathbb{C}^2 \rightarrow \bigotimes_i \mathcal{H}_i$ be the isometries $U|r\rangle = u_r$, $V|r\rangle = v_r$, and set

$$K = (U^\dagger \otimes V^\dagger) R (U \otimes V) \succeq 0,$$

Then G is the $\{|11\rangle, |22\rangle\}$ principal submatrix of K^{Γ_2} , where Γ_2 is the partial transpose on the second auxiliary copy of $\mathbb{C}^2$. With $K_{rs,tu} := \langle r, s | K | t, u \rangle$, this follows from

$$b_3(A_1, A_2) = K_{12,21}.$$

Ordinary Cauchy–Schwarz for K instead gives

$$|b_3(A_1, A_2)|^2 \leq q_3(|u_1\rangle\langle v_2|) q_3(|u_2\rangle\langle v_1|),$$

with the wrong two diagonal factors. For $u_1 = v_1 = |000\rangle$ and $u_2 = v_2 = |111\rangle$, the desired diagonal product is $1/64$, whereas the crossed one is 1 . In the same example the true off-diagonal entry is $(-1/2)^3 = -1/8$, so (32) is saturated. This elementary example rules out a product-switch shortcut.

7.2 Two broad nonnormal closures

The first result constrains one local output–input support. It is independent of the common-plane corollary and covers every tripartite system having at least one qubit-sized local factor.

Theorem 7.2 (Local output–input support overlap). *Let $\xi = \text{vec } C$ have Schmidt rank at most two across the output–input cut. Suppose that, for some site i , there are projections P_i, Q_i such that*

$$\xi = (P_i \otimes \overline{Q_i} \otimes I_{\hat{i}}) \xi, \quad (36)$$

where the overline is entrywise conjugation in the fixed input basis, $\hat{i} = \{1, 2, 3\} \setminus \{i\}$ labels the other two output–input pairs, and $I_{\hat{i}}$ is their identity. If

$$\tau_i = \text{Tr}(P_i Q_i) \leq 2, \quad (37)$$

then $q_3(C) \geq 0$. In particular, this holds if either $\text{rank } P_i \leq 2$ or $\text{rank } Q_i \leq 2$, and hence

$$\min(d_1, d_2, d_3) \leq 2, \quad \text{rank } C \leq 2 \quad \implies \quad q_3(C) \geq 0. \quad (38)$$

Proof. Under vectorization,

$$q_3(C) = \langle \xi, W_3 \xi \rangle, \quad W_3 = \bigotimes_{j=1}^3 W_{d_j}, \quad W_d = I - \frac{1}{2} |\Omega_d\rangle\langle\Omega_d|. \quad (39)$$

Put

$$P = P_i \otimes \overline{Q_i}, \quad \mathcal{K}_i^{\text{supp}} = \text{ran } P_i \otimes \text{ran } \overline{Q_i}.$$

The conjugate on the input support is forced by vectorization:

$$(P_i \otimes \overline{Q_i}) \text{vec } X = \text{vec}(P_i X Q_i).$$

Since $\xi = (P \otimes I_{\hat{\gamma}})\xi$, only the compression of W_{d_i} to $\mathcal{K}_i^{\text{supp}}$ contributes. With

$$|\omega\rangle = P|\Omega_{d_i}\rangle,$$

one has

$$P W_{d_i} P = P - \frac{1}{2} |\omega\rangle\langle\omega| \cong I_{\mathcal{K}_i^{\text{supp}}} - \frac{1}{2} |\omega\rangle\langle\omega| =: G.$$

The maximally entangled contraction identity gives

$$\|\omega\|^2 = \langle \Omega_{d_i} | P_i \otimes \overline{Q_i} | \Omega_{d_i} \rangle = \text{Tr}(P_i^T \overline{Q_i}) = \text{Tr}(P_i Q_i) = \tau_i,$$

where $\text{Tr}(P_i Q_i) = \text{Tr}(P_i Q_i P_i)$ is real and nonnegative. Consequently,

$$\|G - I_{\mathcal{K}_i^{\text{supp}}}\|_2 = \frac{1}{2} \|\omega\|_2 = \frac{\tau_i}{2} \leq 1.$$

Moreover, (37) gives $G \succeq 0$, because its only possibly nonunit eigenvalue is $1 - \tau_i/2$. The Gurvits–Barnum Hilbert–Schmidt ball theorem states that every operator $I_{\mathcal{K}_i^{\text{supp}}} + \Delta \succeq 0$ with $\Delta = \Delta^\dagger$ and $\|\Delta\|_2 \leq 1$ is separable across the bipartition of $\mathcal{K}_i^{\text{supp}}$; the center is the unnormalized identity [54, Theorem 1]. It follows directly that G is separable.

Write a rank-one separable decomposition

$$G = \sum_{\alpha} |a_{\alpha}\rangle\langle a_{\alpha}| \otimes |b_{\alpha}\rangle\langle b_{\alpha}|.$$

Contracting ξ with $\langle a_{\alpha}| \otimes \langle b_{\alpha}|$ leaves a vector ξ_{α} on the other two output–input pairs whose Schmidt rank is still at most two. Consequently,

$$\langle \xi, G \otimes W_{d_j} \otimes W_{d_k} \xi \rangle = \sum_{\alpha} \langle \xi_{\alpha}, W_{d_j} \otimes W_{d_k} \xi_{\alpha} \rangle \geq 0$$

by the two-copy endpoint theorem [29, Theorem A]; see also the concurrent equal-dimension and complementary formulations [26–28]. This proves the first assertion. Since $\text{Tr}(P_i Q_i) \leq \min(\text{rank } P_i, \text{rank } Q_i)$, either support-rank condition implies (37). Taking $P_i = Q_i = I$ when $d_i \leq 2$ proves (38). $\square$

For a singular-value decomposition, let

$$U = \text{span}\{u_1, u_2\}, \quad V = \text{span}\{v_1, v_2\},$$

and let P_i, Q_i be their minimal one-site support projections, equivalently the support projections of $\text{Tr}_{\hat{i}} P_U$ and $\text{Tr}_{\hat{i}} P_V$, where P_U and P_V are the orthogonal projections onto U and V , respectively, and $\text{Tr}_{\hat{i}}$ traces out the two sites other than i . Thus a counterexample, if one exists, must obey

$$\text{Tr}(P_i Q_i) > 2 \quad (i = 1, 2, 3).$$

In particular, both its output and input supports must have dimension at least three at every site. In $3 \times 3 \times 3$, all six local supports would have to be the full qutrit space.

The second closure rests on a switched Schur complement available when one support plane contains a product reference ray. The matrix order in this complement is reversed from the ordinary Schur order. We therefore keep the full co-Choi argument beside the theorem that uses it.

Theorem 7.3 (A product ray in either support plane). *Let $\text{rank } C \leq 2$. If either $\text{ran } C$ or $\text{ran } C^\dagger$ contains a nonzero fully product vector, then $q_3(C) \geq 0$.*

The crucial input is the following product-reference lemma.

Lemma 7.4 (Product-reference switched Schur complement). *Let $u = u_1 \otimes u_2 \otimes u_3$ be a normalized product vector and let x be arbitrary. Compress the positive operator R in (33) on its first copy and define operators on the second copy by*

$$A = (\langle u| \otimes I) R (|u\rangle \otimes I), \quad L = (\langle u| \otimes I) R (|x\rangle \otimes I), \quad D = (\langle x| \otimes I) R (|x\rangle \otimes I).$$

Then $A \succ 0$ and

$$D - LA^{-1}L^\dagger \succeq 0. \quad (40)$$

The ordinary Schur complement of the compression of R contains $L^\dagger A^{-1}L$, whereas the product structure proves the switched order required below.

Proof. Local unitary covariance permits $u_i = |0\rangle$. On one local space put $P = |0\rangle\langle 0|$, $Q = I - P$, and

$$A_0 = I - \frac{1}{2}P, \quad \ell_a = (\langle 0| \otimes I) \left(I - \frac{1}{2}P \right) (|a\rangle \otimes I) = \delta_{a0}I - \frac{1}{2}|a\rangle\langle 0|.$$

For the chosen local basis let $E_{ab} = |a\rangle\langle b|$. Define maps on these matrix units by

$$\Lambda(E_{ab}) = \delta_{ab}I - \frac{1}{2}E_{ab}, \quad \mathcal{M}(E_{ab}) = \ell_a A_0^{-1} \ell_b^\dagger.$$

Writing $x = \sum_a x_a |a\rangle$ and expanding the tensor product gives

$$\begin{aligned} A &= A_{0,1} \otimes A_{0,2} \otimes A_{0,3}, \\ D &= (\Lambda_1 \otimes \Lambda_2 \otimes \Lambda_3)(|x\rangle\langle x|), \\ LA^{-1}L^\dagger &= (\mathcal{M}_1 \otimes \mathcal{M}_2 \otimes \mathcal{M}_3)(|x\rangle\langle x|). \end{aligned} \quad (41)$$

For a linear map $\mathcal{E}$, let

$$J(\mathcal{E}) = \sum_{a,b} E_{ab} \otimes \mathcal{E}(E_{ab}), \quad \hat{J}(\mathcal{E}) = \sum_{a,b} E_{ba} \otimes \mathcal{E}(E_{ab}).$$

If T denotes matrix transposition in the chosen basis, then

$$\hat{J}(\mathcal{E}) = J(\mathcal{E} \circ \mathsf{T}). \quad (42)$$

Thus $\hat{J}(\mathcal{E}) \succeq 0$ is equivalent to complete copositivity of $\mathcal{E}$, meaning that $\mathcal{E} \circ \mathsf{T}$ is completely positive. Set

$$C_0 = \hat{J}(\Lambda) = I - \frac{1}{2}F, \quad S = \hat{J}(\mathcal{M}).$$

The mutually orthogonal spaces $P \otimes P$, $\text{span}\{|0a\rangle, |a0\rangle\}$ for $a \neq 0$, and $Q \otimes Q$ reduce both matrices. On the first space $C_0 = S = 1/2$; on the middle spaces, in the displayed ordered basis,

$$C_0 = \begin{pmatrix} 1 & -1/2 \\ -1/2 & 1 \end{pmatrix}, \quad S = \begin{pmatrix} 1 & -1/2 \\ -1/2 & 0 \end{pmatrix};$$

and on $Q \otimes Q$,

$$C_0 = I - \frac{1}{2}F_Q, \quad S = \frac{1}{2}F_Q.$$

Here F_Q is the swap restricted to $\text{ran } Q \otimes \text{ran } Q$. Thus $C_0 - S \succeq 0$ and $C_0 + S \succeq 0$ on every reducing space, or

$$-C_0 \preceq S \preceq C_0. \quad (43)$$

Since $C_0 \succ 0$, the Hermitian matrix $R = C_0^{-1/2} S C_0^{-1/2}$ is a contraction. After the canonical unitary permutation that groups like tensor factors,

$$\hat{J}\left(\bigotimes_i \Lambda_i - \bigotimes_i \mathcal{M}_i\right) = \left(\bigotimes_i C_{0,i}^{1/2}\right) \left(I - \bigotimes_i R_i\right) \left(\bigotimes_i C_{0,i}^{1/2}\right) \succeq 0.$$

The subscript i here denotes the copy of each one-site object $\Lambda, \mathcal{M}, C_0, R, A_0$ acting at site i . Indeed, every eigenvalue of $\bigotimes_i R_i$ lies in $[-1, 1]$. By (42), the map in parentheses is completely copositive and therefore positive. Applying it to $|x\rangle\langle x|$ in (41) proves (40). $\square$

Proof of Theorem 7.3. Suppose first that $\text{rank } C = 2$ and the range contains a product unit vector u . Complete it to an orthonormal basis u, x of the range and write

$$C = |u\rangle\langle v| + |x\rangle\langle y|,$$

where v, y need not be orthogonal. For this u, x , define

$$A = (\langle u| \otimes I) R (|u\rangle \otimes I), \quad L = (\langle u| \otimes I) R (|x\rangle \otimes I), \quad D = (\langle x| \otimes I) R (|x\rangle \otimes I).$$

Lemma 7.4 gives $D - LA^{-1}L^\dagger \succeq 0$. Set

$$a = \langle v, Av \rangle, \quad d = \langle y, Dy \rangle, \quad g = \langle y, Lv \rangle.$$

Here a and d are the two diagonal values of q_3 , and g is their polarized cross term. Cauchy–Schwarz and (40) give

$$|g|^2 \leq a \langle y, LA^{-1}L^\dagger y \rangle \leq ad.$$

Consequently,

$$q_3(C) = a + d + 2 \text{Re } g \geq (\sqrt{a} - \sqrt{d})^2 \geq 0.$$

The assertion for a product ray in $\text{ran } C^\dagger$ follows from $q_3(C^\dagger) = q_3(C)$; the rank-one case follows by setting one covector to zero. $\square$

The product-ray and local-overlap closures are independent. For example, in $(\mathbb{C}^3)^{\otimes 3}$, let

$$u = |000\rangle, \quad x = \frac{|012\rangle + |120\rangle + |201\rangle}{\sqrt{3}}, \quad v = \frac{|000\rangle + |111\rangle + |222\rangle}{\sqrt{3}}, \quad y = x,$$

and $C = |u\rangle\langle v| + |x\rangle\langle y|$. All one-site supports are the full qutrit space, while $\dim(\text{ran } C + \text{ran } C^\dagger) = 3$ and $CC^\dagger \neq C^\dagger C$. Thus neither the local-overlap theorem nor the common-plane corollary applies, whereas Theorem 7.3 does.

Two more specialized closures—a local-traceless face and operators between distinct locally orthogonal product codes—are proved in Appendix C.

8 Sharpness and the rank boundary

The following constructions separate two issues: optimality of the constants within rank two and necessity of the rank-two hypothesis.

8.1 A nonnormal rank-two/rank-three boundary family

We continue to write $E_{ab} = |a\rangle\langle b|$ for matrix units.

Proposition 8.1 (Nonnormal equality at rank two and negativity at rank three). *Let $m \geq 2$, assume the local dimensions contain a $2 \times 2 \times m$ subspace, and define*

$$C_m = E_{01} \otimes E_{01} \otimes \frac{I_m}{\sqrt{m}}. \quad (44)$$

Then C_m is nonnormal, $\text{rank } C_m = m$, $\|C_m\|_2 = 1$, and

$$q_3(C_m) = 1 - \frac{m}{2}. \quad (45)$$

In particular, C_2 is a genuinely nonnormal rank-two equality operator, whereas $q_3(C_3) = -1/2$.

Proof. For product operators the form factorizes:

$$q_3(X_1 \otimes X_2 \otimes X_3) = \prod_{i=1}^3 \left(\|X_i\|_2^2 - \frac{1}{2} |\text{Tr } X_i|^2 \right).$$

The first two factors of (44) have Hilbert–Schmidt norm one and trace zero, while $I_m/\sqrt{m}$ has Hilbert–Schmidt norm one and trace $\sqrt{m}$. This proves (45). The rank and norm statements are immediate. Finally,

$$C_m C_m^\dagger = E_{00} \otimes E_{00} \otimes \frac{I_m}{m}, \quad C_m^\dagger C_m = E_{11} \otimes E_{11} \otimes \frac{I_m}{m},$$

so C_m is nonnormal. □

8.2 Three-qubit anti-state equality

On one qubit, let K denote conjugation in the computational basis, put

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad J = i\sigma_y,$$

and define the antiunitary $\vartheta = JK$. If K_{global} denotes component-wise conjugation of the full three-qubit coefficient vector, set

$$\Theta = J^{\otimes 3} K_{\text{global}} = \vartheta^{\otimes 3}.$$

Here the tensor product of local antiunitaries is understood canonically. The local map $\vartheta = i\sigma_y K$ is the spin flip entering concurrence and universal state inversion [55, 56]; its odd-party tensor power is the anti-state map used in [57].

Proposition 8.2 (Anti-state saturation). *For every normalized three-qubit u , let $v = \Theta u$. Then*

$$u \perp v, \quad 3p_3(u, v) - p_2(u, v) = \sqrt{a(u)a(v)}.$$

Moreover, for all $\lambda, \mu \geq 0$,

$$q_3(\lambda P_u + \mu P_v) = (\lambda - \mu)^2 \left(\frac{1}{8} + a(u) \right). \quad (46)$$

Proof. Since $\vartheta^2 = -I$, one has $\Theta^2 = -I$, and Kramers orthogonality gives $\langle u, \Theta u \rangle = 0$. Each local antisymmetric two-qubit space is spanned by the singlet. Contracting $u \otimes \Theta u$ with the three normalized singlets gives

$$\langle \omega |^{\otimes 3} (u \otimes \Theta u) = -\frac{1}{2\sqrt{2}}, \quad |\omega\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}},$$

where the phase depends on the convention for ϑ , but the modulus does not. Indeed, the one-qubit contraction is $\langle \omega | (x \otimes \vartheta y) = -\langle y, x \rangle / \sqrt{2}$. Consequently,

$$p_3(u, \Theta u) = \frac{1}{8}.$$

The spin flip maps each one-qubit reduction to $\rho_i^v = I - \rho_i^u$. Explicitly, if

$$\rho = \begin{pmatrix} a & c \\ \bar{c} & 1-a \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

then

$$J\bar{\rho}J^\dagger = \begin{pmatrix} 1-a & -c \\ -\bar{c} & a \end{pmatrix} = I - \rho.$$

Consequently,

$$\sum_i \text{Tr}(\rho_i^u \rho_i^v) = \sum_i \left(1 - \text{Tr}[(\rho_i^u)^2] \right) = 4a(u).$$

Equation (19), together with $p_3 = 1/8$, gives

$$p_2 = \frac{3}{8} - a(u), \quad a(v) = a(u).$$

Therefore $3p_3 - p_2 = a(u)$, proving equality in (9). Substituting the same identities into (22) and then (21) proves (46). $\square$

Thus every balanced anti-state mixture is an equality state, and the constant in Theorem 1.2 cannot be reduced. For the product antipodes $u = |000\rangle$, $v = |111\rangle$, one has $a(u) = a(v) = 0$, and (46) reduces to

$$q_3(\lambda P_u + \mu P_v) = \frac{(\lambda - \mu)^2}{8}.$$

This attains equality in the positive lower bound (11), so its coefficient is sharp as well.

8.3 Failure at rank three

Proposition 8.3 (Rank-three counterexample). *There is a positive tripartite state C of rank three for which both $\mathcal{S}(C) < 0$ and $q_3(C) < 0$.*

Proof. Let

$$C = |\psi\rangle\langle\psi|_{12} \otimes \tau_3, \quad e = \text{Tr}(\psi_1^2) = \text{Tr}(\psi_2^2), \quad t = \text{Tr} \tau_3^2.$$

Here $\psi_1 = \text{Tr}_2 |\psi\rangle\langle\psi|$ and $\psi_2 = \text{Tr}_1 |\psi\rangle\langle\psi|$, while τ_3 is a density operator on subsystem 3. The global, one-body, and two-body purity sums are

$$\text{Tr} C^2 = t, \quad \sum_i \text{Tr} C_i^2 = 2e + t, \quad \sum_i \text{Tr} C_i^2 = 1 + 2et.$$

Consequently

$$\mathcal{S}(C) = 2(2t - 1)(1 - e). \quad (47)$$

Choose

$$|\Phi_3\rangle = \frac{1}{\sqrt{3}} \sum_{a=0}^2 |aa\rangle,$$

maximally entangled on $\mathbb{C}^3 \otimes \mathbb{C}^3$, and set $\psi = \Phi_3$, $\tau_3 = I_3/3$. Then $e = t = 1/3$, $\text{rank } C = 3$, and

$$\mathcal{S}(C) = -\frac{4}{9}, \quad q_3(C) = \frac{1}{4}\mathcal{S}(C) + \frac{1}{8}(2t - 1) = -\frac{11}{72}.$$

□

This example lies outside the Schmidt-rank-at-most-two witness domain in (2). It blocks an arbitrary-rank extension of Theorem 1.3 and of the associated linear-entropy inequality. Because it is not an admissible rank-two witness, it has no direct implication for Werner-state distillability.

Within the tensor-product family in the proof, rank two is the exact threshold for $\mathcal{S}$: if $\text{rank } \tau \leq 2$, then $t \geq 1/2$, making (47) nonnegative.

9 Operational and physical meaning

Equation (2) identifies C with the coefficient matrix of a Schmidt-rank-two test vector. The main results impose three successive filters on a negative witness at $\alpha = -1/2$: it cannot be normal (including positive semidefinite); its global initial and final two-planes must be distinct and product-ray-free; and their local output–input overlap must exceed two at every site. Appendix C adds two independent exclusions, namely every local-traceless face and the crossed locally orthogonal product-code class. Each restriction concerns the coherent test vector; none assumes that the Werner state has low rank.

The symbol C has two uses. In the distillation criterion it is an arbitrary coefficient matrix and need not be Hermitian. If $C \succeq 0$ and $\text{Tr } C = 1$, the same matrix is also a tripartite density operator. In that interpretation, $\mathcal{S}(C) \geq 0$ is exactly the linear-entropy relation (13). Proposition 8.3 shows that this entropy statement is rank-sensitive and fails as a universal entropy law.

9.1 Local SWAP parity and measurable purities

For density operators ρ, σ on a multipartite system, let F_i exchange the two copies of subsystem i , and define $F_S = \prod_{i \in S} F_i$, with $F_\emptyset = I$. Also write $\rho_S = \text{Tr}_{\bar{S}} \rho$ and $\sigma_S = \text{Tr}_{\bar{S}} \sigma$. The SWAP identity reads

$$\text{Tr}[(\rho \otimes \sigma)F_S] = \text{Tr}(\rho_S \sigma_S). \quad (48)$$

When $\rho = \sigma$, the right side is the subsystem purity $\text{Tr} \rho_S^2$. Controlled-SWAP circuits, two-copy beam-splitter interference, and randomized-measurement protocols provide experimentally established routes to these nonlinear observables [37–42, 58]. Because the local swaps commute, the complete joint parity distribution is obtained from the correlators in (48) by the Walsh transform written explicitly in (17). The quantities a, p_2, p_3 are particular coarse-grainings of this distribution. Antisymmetric sectors themselves also carry nontrivial entanglement-cost and distillable-key bounds [59]; here they enter specifically as outcomes of copy-exchange parity measurements.

For a pure tripartite vector w ,

$$4a(w) = \sum_{i=1}^3 (1 - \text{Tr}[(\rho_i^w)^2]).$$

Each summand is the linear entropy of entanglement across the bipartition $i : \bar{i}$. Theorem 1.2 therefore states that the weighted parity combination $3p_3 - p_2$, comparing the fully antisymmetric outcome with the two-minus sectors, is bounded by the geometric mean of the total one-versus-rest entanglement of the two rays. This gives the Plücker inequality an operational two-copy reading: it is a sharp compatibility constraint on observable local exchange parities.

9.2 Transition coherence and the second compound

The operator

$$X_k = \text{Tr}_k |u\rangle\langle v|$$

is the coherence between two orthogonal rays that remains after subsystem k is discarded. Theorem 1.1 controls the total Hilbert–Schmidt strength of its two further reductions. The quantity retained in the proof,

$$\|X_k\|_1^2 - \|X_k\|_2^2 = 2\|\bigwedge^2 X_k\|_1,$$

is the sum of all pairwise products of singular values. It vanishes exactly when X_k has rank at most one and hence detects the irreducibly two-dimensional part of the reduced coherence. The exterior-square estimate bounds that component by the geometric mean of the two local linear entropies. This is the compensation mechanism that is absent from a treatment of the SWAP-sector weights as unconstrained classical probabilities.

10 Discussion

A three-step dimension-free argument drives the results. The double-antisymmetric projector first gives the exact Schmidt-rank-two overlap $1/2$. The SVD then converts that projection statement into the partial-trace inequality of Theorem 1.1. Finally, the second compound matrices of the three transition operators convert the nuclear-norm excess into local linear entropies. The three cutwise estimates sum exactly to the Plücker combination in (9).

This chain closes every positive semidefinite rank-two test and, after combining the two spectral signs, every normal rank-two test in arbitrary finite local dimensions. The anti-state family proves

that the Plücker constant and the positive-spectrum square are sharp. Proposition 8.1 supplies a genuinely nonnormal rank-two equality and a rank-three negative continuation, while Proposition 8.3 rules out an arbitrary-rank extension of the positive result.

Consequently, any negative rank-two endpoint witness, if exists, must be genuinely nonnormal. Its initial and final two-planes must be distinct and product-ray-free; their local output–input overlap must exceed two at every site; no local partial trace may vanish; and the operator must lie outside the crossed locally orthogonal product-code sector. This necessary profile is the geometric content of the sector theorems.

Beyond normality, Proposition 7.1 reduces the unrestricted question to one crossed-Gram determinant. The local support and product-ray theorems remove two broad geometric regions. Even a future closure of the complete three-copy problem would remain a finite-copy result: for every prescribed copy number there exist states whose distillation requires more copies [9]. The asymptotic NPT bound-entanglement problem is therefore strictly stronger. The present results nevertheless convert the first unresolved three-copy layer from an unstructured rank-two search into a two-plane compatibility problem with sharp analytic boundaries.

A Vectorization and the Werner distillability form

We record the contraction behind (2). Fix product bases on

$$A = A_1 \otimes \cdots \otimes A_k, \quad B = B_1 \otimes \cdots \otimes B_k,$$

with $A_j \simeq B_j \simeq \mathbb{C}^d$. For $J \subseteq [k] = \{1, \dots, k\}$, let

$$R_J = \bigotimes_{j \in J} |\Omega_d\rangle\langle\Omega_d|_{A_j B_j} \bigotimes_{j \notin J} I_{A_j B_j}.$$

A direct index contraction gives

$$\langle \text{vec } C | R_J | \text{vec } C \rangle = \| \text{Tr}_J C \|_2^2. \quad (49)$$

Indeed, each maximally entangled projector identifies the row and column index at the corresponding tensor factor and sums over that common index; the remaining free indices are exactly the matrix indices of $\text{Tr}_J C$. The cases $J = \emptyset$ and $J = [k]$ reduce respectively to $\|C\|_2^2$ and $|\text{Tr } C|^2$.

Since

$$(\rho_\alpha^{(d)})^\Gamma = \frac{I + \alpha |\Omega_d\rangle\langle\Omega_d|}{d^2 + \alpha d},$$

expanding its k -fold tensor power and applying (49) proves (2). Finally, the standard distillability theorem states that ρ is k -copy distillable exactly when $(\rho^\Gamma)^{\otimes k}$ has negative expectation on a vector of Schmidt rank at most two [2–4]. Vectorization converts that Schmidt-rank condition into $\text{rank } C \leq 2$.

B Swap and transition identities

For $S \subseteq [k]$, let $F_S = \prod_{i \in S} F_i$ be the product of the local swaps between the two copies, with $F_\emptyset = I$, and let $\bar{S} = S^c = [k] \setminus S$. For operators on a tensor product, the swap trick gives

$$\text{Tr}[(A \otimes B)F_S] = \text{Tr}[(\text{Tr}_{\bar{S}} A)(\text{Tr}_{\bar{S}} B)]$$

in its bilinear form, whereas the Hilbert–Schmidt form is

$$\mathrm{Tr}[(A^\dagger \otimes B)F_S] = \mathrm{Tr}[(\mathrm{Tr}_{\bar{S}} A)^\dagger \mathrm{Tr}_{\bar{S}} B].$$

Consequently, for an arbitrary operator C ,

$$\|\mathrm{Tr}_J C\|_2^2 = \mathrm{Tr}[(C^\dagger \otimes C)F_{J^c}].$$

Hence, for every tripartite C ,

$$q_3(C) = \mathrm{Tr} \left[(C^\dagger \otimes C) \prod_{i=1}^3 \left(F_i - \frac{1}{2} I \right) \right].$$

For Hermitian C , $C^\dagger \otimes C = C \otimes C$.

For normalized u, v ,

$$\|\mathrm{Tr}_S |u\rangle\langle v|\|_2^2 = \mathrm{Tr}(\rho_S^u \rho_S^v).$$

Expanding $\prod_i (I \pm F_i)/2$, using $\langle u, v \rangle = 0$, and grouping the sectors with exactly two or three minus signs yields (19). Repeating the expansion for $u \otimes u$, for which only even local parity survives, yields (16).

For completeness, the kernel identity (25) follows term by term from

$$\begin{aligned} \langle \mathrm{Tr}_{\mathcal{K}} |u\rangle\langle v|, \mathrm{Tr}_{\mathcal{K}} |s\rangle\langle t| \rangle_{\mathrm{HS}} &= \langle u \otimes t, F_{\mathcal{L}}(v \otimes s) \rangle, \\ \langle \mathrm{Tr}_{\mathcal{L}} |u\rangle\langle v|, \mathrm{Tr}_{\mathcal{L}} |s\rangle\langle t| \rangle_{\mathrm{HS}} &= \langle u \otimes t, F_{\mathcal{K}}(v \otimes s) \rangle, \\ \overline{\mathrm{Tr} |u\rangle\langle v|} \mathrm{Tr} |s\rangle\langle t| &= \langle u \otimes t, v \otimes s \rangle. \end{aligned}$$

C Additional closed nonnormal sectors

The following two results are independent of the main nonnormal closures.

C.1 The local-traceless sector

For $S \subseteq \{1, 2, 3\}$, let

$$q_S(X) = \left\langle X, \prod_{i \in S} \left(\mathrm{id} - \frac{1}{2} \mathrm{Tr}_i^* \mathrm{Tr}_i \right) X \right\rangle_{\mathrm{HS}},$$

where Tr_i^* is the Hilbert–Schmidt adjoint of Tr_i . Explicitly, up to the canonical permutation that restores the original factor order, $\mathrm{Tr}_i^*(Y) = I_i \otimes Y$. Thus $q_{\{1,2,3\}} = q_3$, while a two-element S gives the two-copy endpoint form on those sites, acting blockwise on the spectator factor.

Theorem C.1 (Local-traceless rank-two sector). *If $\mathrm{rank} C \leq 2$ and $\mathrm{Tr}_i C = 0$ for at least one site i , then $q_3(C) \geq 0$.*

Proof. If $\mathrm{Tr}_i C = 0$, then $(\mathrm{id} - \frac{1}{2} \mathrm{Tr}_i^* \mathrm{Tr}_i)C = C$. Hence

$$q_3(C) = q_{\{1,2,3\} \setminus \{i\}}(C).$$

Choose an orthonormal basis $\{|a\rangle\}$ at the spectator site i , and define the explicit block compressions

$$C_{ab} = (\langle a| \otimes I)C(|b\rangle \otimes I).$$

The two-copy form acts blockwise, and the matrix units $|a\rangle\langle b|$ are Hilbert–Schmidt orthonormal, so

$$q_{\{1,2,3\} \setminus \{i\}}(C) = \sum_{a,b} q_{\{1,2,3\} \setminus \{i\}}(C_{ab}).$$

Each C_{ab} is a compression of C , hence $\mathrm{rank} C_{ab} \leq \mathrm{rank} C \leq 2$. Every summand is nonnegative by the arbitrary-dimension two-copy endpoint theorem [29, Theorem A]. $\square$

C.2 Crossed locally orthogonal product codes

Theorem C.2 (Crossed locally orthogonal code sector). *Let $V : \mathbb{C}^m \rightarrow \bigotimes_{i=1}^3 \mathcal{H}_i$ and $W : \mathbb{C}^n \rightarrow \bigotimes_{i=1}^3 \mathcal{H}_i$ be the isometries*

$$V|r\rangle = \bigotimes_{i=1}^3 |e_r^{(i)}\rangle, \quad W|s\rangle = \bigotimes_{i=1}^3 |f_s^{(i)}\rangle,$$

where each family $\{e_r^{(i)}\}_{r=1}^m$ and $\{f_s^{(i)}\}_{s=1}^n$ is orthonormal at every site i . No relation between the e - and f -families is assumed. Put

$$S_i(r, s) = \langle f_s^{(i)}, e_r^{(i)} \rangle, \quad T = S_1 \circ S_2 \circ S_3,$$

where $\circ$ is the Hadamard product, and

$$w_{rs} = 1 - \frac{1}{2} \sum_i |S_i(r, s)|^2 + \frac{1}{4} \sum_{i < j} |S_i(r, s) S_j(r, s)|^2.$$

Then, for every $M \in \mathbb{C}^{m \times n}$,

$$q_3(VMW^\dagger) = \sum_{r,s} w_{rs} |M_{rs}|^2 - \frac{1}{8} \left| \sum_{r,s} T_{rs} M_{rs} \right|^2. \quad (50)$$

Consequently,

$$\text{rank } M \leq 2 \implies q_3(VMW^\dagger) \geq \frac{2 - \sigma_1(T)^2 - \sigma_2(T)^2}{8} \|M\|_2^2 \geq 0. \quad (51)$$

Here $\sigma_1(T) \geq \sigma_2(T)$ are the two largest singular values, and $\sigma_2(T) = 0$ when T has fewer than two singular values.

Proof. Expanding $C = VMW^\dagger$ gives

$$C = \sum_{r,s} M_{rs} \bigotimes_{i=1}^3 |e_r^{(i)}\rangle \langle f_s^{(i)}|.$$

For every proper subset $J \subsetneq \{1, 2, 3\}$,

$$\text{Tr}_J C = \sum_{r,s} M_{rs} \left(\prod_{i \in J} S_i(r, s) \right) \bigotimes_{k \notin J} |e_r^{(k)}\rangle \langle f_s^{(k)}|.$$

Since $J^c \neq \emptyset$, orthonormality at any site $k \in J^c$ makes the remaining tensor factors Hilbert–Schmidt orthogonal for different ordered pairs (r, s) . Therefore

$$\|\text{Tr}_J C\|_2^2 = \sum_{r,s} |M_{rs}|^2 \prod_{i \in J} |S_i(r, s)|^2 \quad (J \subsetneq \{1, 2, 3\}).$$

On the other hand,

$$\text{Tr } C = \sum_{r,s} M_{rs} \prod_i S_i(r, s) = \sum_{r,s} T_{rs} M_{rs}.$$

Substitution in (3) proves (50).

For $0 \leq a_i \leq 1$, the multiaffine polynomial

$$1 - \frac{1}{2} \sum_i a_i + \frac{1}{4} \sum_{i < j} a_i a_j$$

has minimum $1/4$ on $[0, 1]^3$, as follows by checking its eight vertices. Hence $w_{rs} \geq 1/4$.

If $\mathbf{E}_i, \mathbf{G}_i$ are the isometries with columns $e_r^{(i)}$ and $f_s^{(i)}$, then $S_i = \overline{\mathbf{E}_i^\dagger \mathbf{G}_i}$, so every S_i is a contraction. With diagonal isometries $\Delta_k |r\rangle = |rrr\rangle$,

$$T = \Delta_m^\dagger (S_1 \otimes S_2 \otimes S_3) \Delta_n,$$

and hence $\|T\|_\infty \leq 1$. Finally,

$$\sum_{r,s} T_{rs} M_{rs} = \langle \overline{T}, M \rangle_{\text{HS}}.$$

If $\text{rank } M \leq 2$, von Neumann's trace inequality gives

$$|\langle \overline{T}, M \rangle_{\text{HS}}|^2 \leq (\sigma_1(T)^2 + \sigma_2(T)^2) \|M\|_2^2.$$

Combining the last three estimates with (50) proves (51). $\square$

Remark C.3. Taking $V = W$ and the same local bases gives $S_i = T = I_m$, $w_{rr} = 1/4$, and $w_{rs} = 1$ for $r \neq s$. Formula (50) then becomes

$$q_3(VMV^\dagger) = \|M\|_2^2 - \frac{3}{4} \sum_r |M_{rr}|^2 - \frac{1}{8} |\text{Tr } M|^2,$$

which is the same-code locally orthogonal formula.

Remark C.4 (A fully supported square-zero example). In $3 \times 3 \times 3$, with labels understood modulo three, take

$$V|r\rangle = |r, r, r\rangle, \quad W|s\rangle = |s, s+1, s+2\rangle$$

and

$$M = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & -1 \end{pmatrix}^T.$$

The two code spaces are orthogonal, $\text{rank } M = 2$, and the row and column two-planes of M have nonzero weight on all three labels. Thus $C = VMW^\dagger$ is square zero, genuinely nonnormal, and has full output and input support at every local site. In a locally orthogonal code, a fully product vector is a single codeword, and all three local partial traces of C are nonzero. Hence none of the preceding closed-sector theorems applies. Here exactly one of the three local overlaps is nonzero for each ordered pair (r, s) , so $w_{rs} = 1/2$, $T = 0$, and

$$q_3(C) = \frac{1}{2} \|M\|_2^2 = 3.$$

Data availability

Data sharing is not applicable to this article as no datasets were generated or analysed during the current study.

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